Autonomous Production Choke Controller

for a Single Naturally Flowing Oil Well

Brute-force Model Predictive Control · control interval Ts = 1 hour

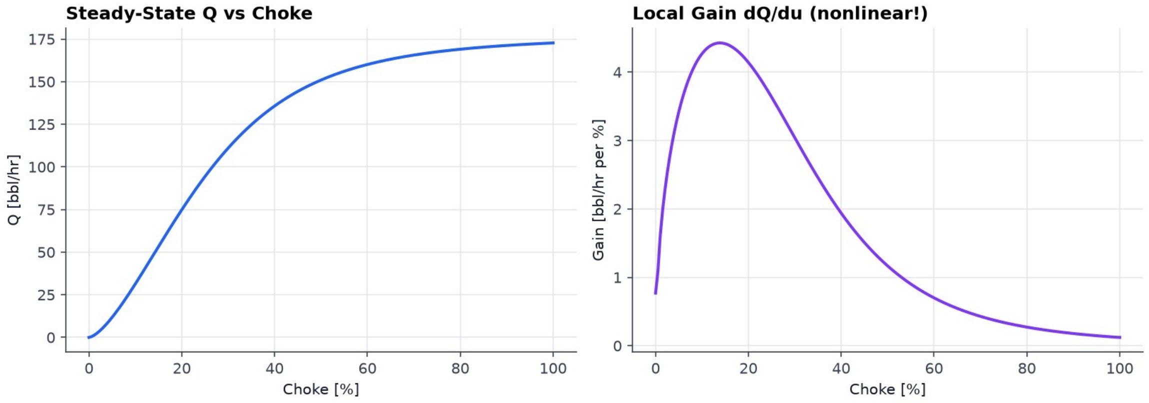
Maximises safe production; refuses targets that cannot be met safely

0 constraint violations across 300 closed-loop runs | targets tracked to <1 % | 98.6 % of max safe rate when target infeasible

PROCESS UNDERSTANDING & MODEL

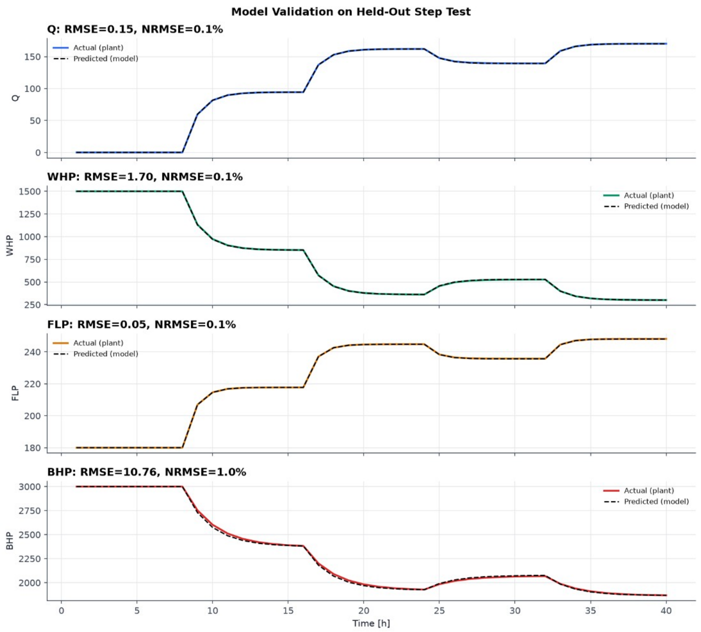
Step-test results (our own experiments)

- 15 choke steps, up and down, 5–20 % magnitude, each held 8 h (≈4× slowest τ) to reach true steady state
- Opening the choke raises Q and FLP, lowers WHP and BHP — one input, four strongly coupled outputs
- Gain dQ/du collapses 3.91 → 0.15 bbl/hr per % across the range (~26×) → the process is strongly nonlinear
- Provided reference dataset used for illustration only — never for fitting, per the problem statement



Model assumptions

- Linear IPR (constant productivity index); turbulent tubing friction $\propto Q^2$; choke orifice $Q = C_v \cdot f(u) \cdot \sqrt{WHP - FLP}$
- One first-order lag per output; dead time negligible ($\theta \le 0.09$ h)
- WHT and AP monitored and logged, but not active constraints



Dynamic model developed

- $$y(k+1) = y(k) + \alpha \cdot \left(y_{ss}(u) + d - y(k) \right), \quad \alpha = 1 - e^{-T_s/\tau}$$
- Nonlinearity captured as a piecewise-linear steady-state curve $y_{ss}(u)$; dynamics stay linear → simple and explainable
- Fitted τ: Q 0.99 h · WHP 1.18 h · FLP 0.80 h · BHP 1.75 h
- **Validated on HELD-OUT steps: NRMSE 0.09 % (Q), 0.95 % (BHP)**

TECHNICAL APPROACH — CONTROL STRATEGY

Prediction methodology

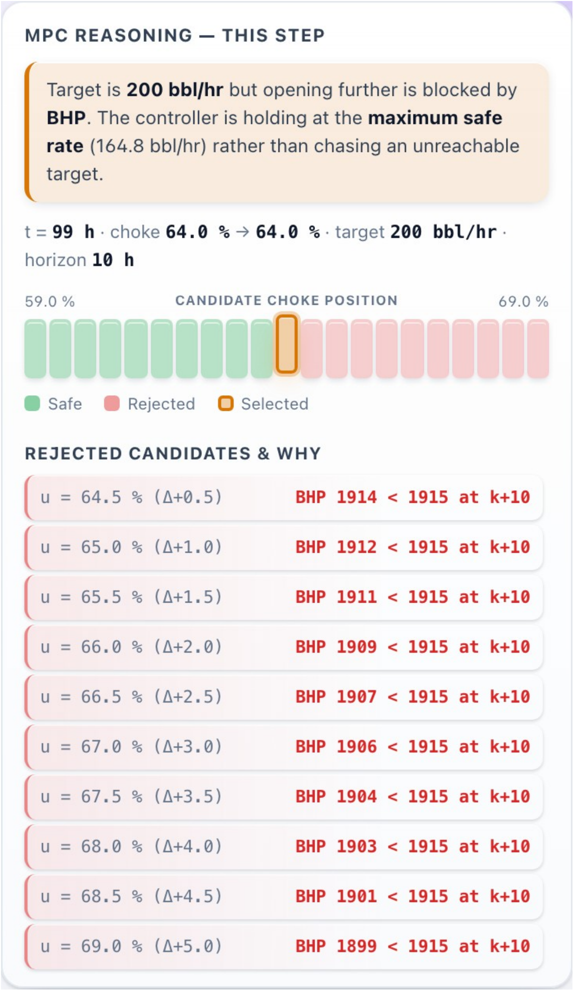
- Enumerate ALL candidate moves within the ramp limit: $u \pm 5$ % on a 0.5 % grid → 21 candidates
- Roll each forward 10 h through the identified model (move-blocked: apply, then hold)
- Horizon must outrun the slowest lag: BHP $\tau = 1.75$ h, so 10 h ≈ $5.7\,\tau$
- Every prediction restarts from the current measurement → model error never compounds
- Measurement filter + bias correction remove hunting and steady-state offset

Choke move selection logic

- **$J = (Q\_pred\_end - Q\_target)^2 + 0.05 \cdot (\Delta u)^2$ → closest to target, penalise large moves**
- Deterministic tie-breaks: lowest cost → smallest $|\Delta u|$ → lower choke (conservative)
- Deadband: move only if it buys ≥ 2 $(\text{bbl/hr})^2$ improvement → zero valve chatter

Constraint handling

- Reject any candidate whose predicted WHP/FLP/BHP breaches limits at ANY point in the horizon
- Safety margins inside each hard limit (10/8/15 psi) cover sensor noise + model error + filter lag
- If ALL candidates unsafe → pick the least-bad (smallest predicted violation), never lurch
- Infeasible target needs no special case: 'open further' is filtered out every step → the controller pins itself at the safe boundary



Live MPC reasoning: every candidate, why each was rejected

FEASIBILITY AND VIABILITY

300 closed-loop runs	0 constraint violations	98.6 % of max safe rate	19/19 tests passing

Feasibility — proven, not asserted

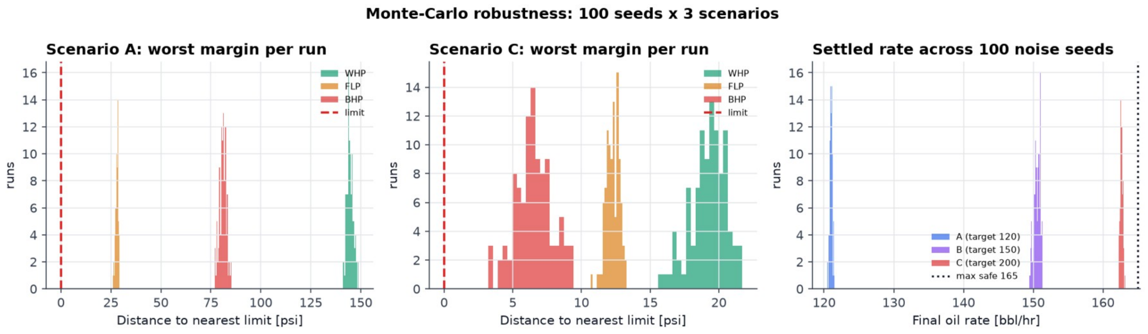
- Monte-Carlo: 100 independent noise seeds × 3 scenarios; envelope checked on every sample
- Worst margin to any limit: A +26 psi · B +16 psi · C +3.2 psi
- Runs end-to-end in ~2 min; no GPU, no internet, no solver library

Challenges found — and fixed

- Horizon shorter than the slowest lag → BHP kept falling after the horizon (11/100 runs breached)
- Coarse step-test knots → chord across a convex curve over-estimated BHP by 8 psi (unsafe direction)
- **Both passed single-run testing; only the Monte-Carlo exposed them**

Risk strategy

- Margins sized for noise + model error + filter lag, and re-verified over 300 runs
- Cost of safety quantified, not hidden: 2.3 bbl/hr (1.4 %) below the theoretical max
- Controller reads ONLY step() outputs → official simulator drops in with zero changes (proven by a test against a different plant)



ARTIFACTS — RESULTS

Scenario outcomes & tracking performance

A — Startup to target: settled 121.0 ± 0.6 bbl/hr (target 120) · choke 33.5 % · min BHP 2184 psi · 0.8 % error

B — Target tracking 100→150: settled 150.8 ± 0.8 bbl/hr (target 150) · choke 50.0 % · min BHP 1989 psi · 0.5 % error

C — Infeasible target 200: settled 162.7 ± 0.9 bbl/hr (target 200) · choke 64.0 % · min BHP 1906 psi · capped at max safe rate

Safety performance

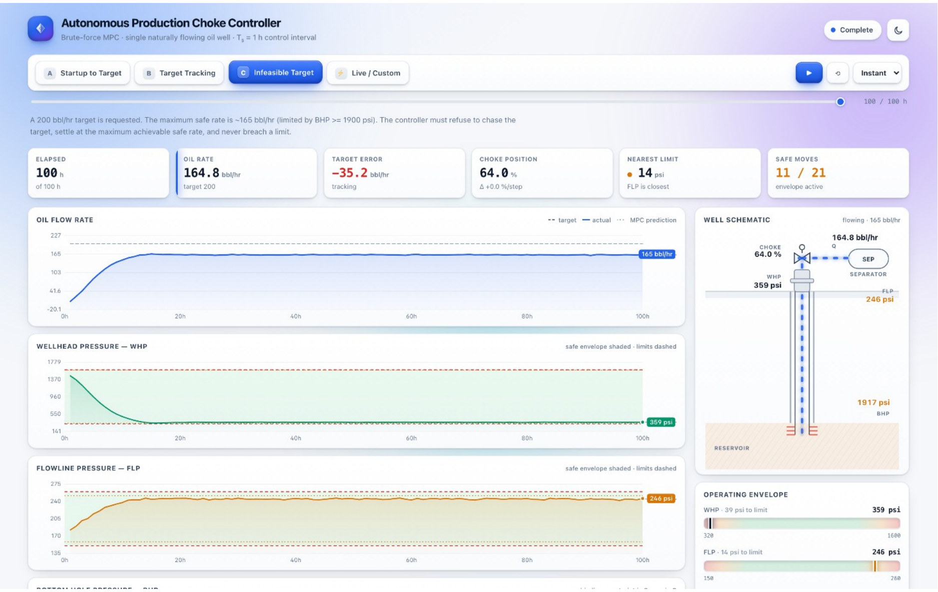
WHP ≥ 320, FLP ≤ 260, BHP ≥ 1900 psi, |Δu| ≤ 5 %/step — held on EVERY sample

Automated PASS/FAIL audit in the pipeline; non-zero exit if any check fails

Deliverables

Simulator · step tests · model ID · MPC · 3 scenarios · Monte-Carlo · 19 tests · dashboard · report

One command reproduces everything: python run_all.py



All three scenarios: target tracked when feasible (A, B); refused and capped at the safe maximum when not (C) — BHP rests on its limit without ever crossing it.

RESEARCH AND REFERENCES

Lessons learned

- Where you place step-test knots is a SAFETY decision — linear interpolation across a convex curve is optimistic exactly where the constraint binds
- A horizon shorter than the slowest time constant is not conservative, it is unsafe
- One good run is not evidence; a safety claim about a stochastic system needs a distribution
- Margins must cover estimator lag and model error, not just sensor noise
- Every filter is a trade: ours removed constraint hunting but cost response lag that had to be paid for in margin
- Infeasibility needs no special-case code — it falls out of predict → filter → select

Engineering references

- Inflow performance / linear IPR (constant productivity index) — standard above bubble point
- Choke orifice / valve-sizing relation $Q = Cv \cdot f(u) \cdot \sqrt{\Delta P}$
- Dynamic Matrix Control: move blocking, output disturbance (bias) estimation, constraint back-off
- Open-loop step testing and FOPDT identification for control-oriented models

Provided material used

- Autonomous_Choke_Control_Simulated_Dataset.csv — reference only, as the problem statement directs; never used to fit the model
- Compared against our physics stand-in and the differences reported openly (reference is near-linear; its FLP falls with rate)

Project artifacts

- src/ well_simulator · model · controller · simulator_interface
- scripts/ reference EDA · step tests · model ID · scenarios · robustness · capture · PDF
- tests/ 19 tests incl. a swap-in proof against a different plant
- dashboard/ self-contained HTML/JS demo (no build, no server needed)
- report/ENGINEERING_REPORT.md → submission/ENGINEERING_REPORT.pdf

Note on the simulator

The simulator module itself was not supplied, so we built a documented physics-based stand-in from the process description. The controller touches only step()/reset(), so the official simulator substitutes with zero changes.